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Paper Code : BS-M301 Mathematics III

UPID : 003465

CS/B.TECH(N)/ODD/SEM-3/3465/2025-2026/1059

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[1 × 10 = 10]

(i) Write the value of kurtosis for a normal distribution.

(ii) Write the auxiliary equations for the first-order PDE

$$y u_x + x u_y = 0$$

(iii) Write the Laplacian of u in two-dimensional Cartesian coordinates.

(iv) For a continuous random variable with density $f(x) = 2x$, $0 < x < 1$, find $P(X < 1/2)$.

(v) Find the mean of the data: 2, 4, 6, 8

(vi) Write the test statistic for large sample test for difference of means.

(vii) For the heat equation

$$u_t = \alpha^2 u_{xx}$$

state the separated form of solution.

(viii) Two coins are tossed. Find the probability of getting exactly one head.

(ix) Find Spearman's rank correlation coefficient if $\sum d^2 = 12$ and $n = 5$.

(x) If $P(A) = 0.4$ and $P(B) = 0.5$ with A, B independent, find $P(A \cap B)$.

(xi) Identify the type of PDE:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

(xii) If $X \sim N(0, 1)$, write $P(|X| < 1)$ in standard normal notation.

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 × 3 = 15]

2. Find the probability that a binomial random variable with $n = 10$, $p = 0.1$ takes the value $X = 0$.

[5]

3. Solve the first-order linear PDE

[5]

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = z$$

4. Test the significance of a single proportion:

[5]

In a sample of 400 items 220 are defective. Test at 5% level whether $p = 0.5$.

5. Solve the wave equation using D'Alembert's formula:

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2}$$

given

$$u(x, 0) = x^2, \quad \frac{\partial u}{\partial t}(x, 0) = 0$$

6. Fit a curve of the form

$$y = ax^2 + bx + c$$

by the method of least squares to the data:

x	-2	-1	0	1	2
y	9	4	1	4	9

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

7. (a) In a certain factory, machine A produces 60% of the items and machine B produces 40%. The probability that an item produced by machine A is defective is 0.02, while the probability that an item produced by machine B is defective is 0.05.

[5]

If an item selected at random is found to be defective, find the probability that it was produced by machine A.

- (b) A discrete random variable X has the following probability distribution:

[5]

X	0	1	2	3
P(X)	0.1	0.3	0.4	0.2

Find

(i) $E(X)$

(ii) $\text{Var}(X)$

- (c) Let X and Y be independent random variables such that:

[5]

$$E(X) = 2, \quad \text{Var}(X) = 3$$

$$E(Y) = 1, \quad \text{Var}(Y) = 2$$

Find

(i) $E(X + Y)$

(ii) $\text{Var}(X + Y)$

8. (a) Solve the radial equation:

[7]

$$r^2 \frac{d^2 R}{dr^2} + r \frac{dR}{dr} + (\lambda r^2 - 1)R = 0$$

(b)

[8]

Solve the diffusion equation:

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$$

subject to

$$u(0, t) = 0, \quad u(\pi, t) = 0$$

and

$$u(x, 0) = 3 \sin x$$

9. (a) The lifetime (in hours) of a bulb follows an exponential distribution with probability density function: [5]

$$f(x) = \lambda e^{-\lambda x}, \quad x > 0$$

where $\lambda = 0.5$.

Find the probability that the bulb lasts more than 3 hours.

- (b) The joint probability density function of X and Y is: [5]

$$f(x, y) = 2, \quad 0 < x < y < 1$$

Find

(i) the marginal density of X ,

(ii) $P(X < 0.5)$

- (c) A factory produces bolts, and 2% of them are defective. If 200 bolts are selected at random, find the probability that exactly 3 bolts are defective using Poisson approximation. [5]

10. (a) Using Duhamel's principle, find the solution of: [7]

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

with

$$u(x, 0) = 0, \quad \frac{\partial u}{\partial t}(x, 0) = x$$

- (b) Find the Laplacian of the function: [8]

$$\phi(r, \theta, z) = r^2 \cos \theta$$

in cylindrical polar coordinates.

11. (a) A sample of 64 students has mean marks 72, population standard deviation 8. Test if population mean is 70 at 5% level. <https://www.makaut.com> [5]

- (b) The regression coefficient of Y on X is $b_{yx} = 0.6$ and the regression coefficient of X on Y is $b_{xy} = 0.4$. [5]

Find the correlation coefficient between X and Y and comment on its nature.

- (c) The following table shows the distribution of gender and preference for online classes: [5]

	Like	Dislike	Total
Male	30	20	50
Female	20	30	50
Total	50	50	100

Test at 5% level whether gender and preference are independent